Bennu's returned samples recorded a solar nebula magnetic field

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Abstract: Magnetic fields in protoplanetary disks are thought to drive angular momentum transport and mass accretion. Paleomagnetic measurements of meteorites support the central role of magnetic fields in the early solar system by preserving thermally acquired magnetizations recorded during their formation. An additional mechanism to probe the solar nebula magnetism is that of chemical remanent magnetization, acquired when magnetic carriers form during parent body aqueous alteration, recording the ambient nebular field at the time and location of alteration. Here, we report paleomagnetic and rock magnetic measurements of subsamples from asteroid 101955 Bennu, returned by NASA's OSIRIS-REx mission. We report magnetic fields >5 μT in the Bennu formation region, likely beyond the carbonaceous chondrite reservoir carried by single domain to vortex state magnetite and pyrrhotite. Placing Bennu's field value in its inferred formation location, alongside other inner and outer solar system records, supports a field strength that decreases with heliocentric distance, consistent with magnetohydrodynamic models of magnetically driven disk accretion. In contrast to Ryugu, a CI-like chondrite similar to Bennu that lacks high-coercivity magnetic records despite exhibiting comparable magnetic-carrier morphologies, Bennu preserves evidence of distinct volatile abundances, suggesting divergent aqueous alteration pathways on their parent bodies and highlighting the influence of alteration history on the acquisition of magnetic records in the early solar system.

Main Text

Introduction

Magnetic fields are thought to play a fundamental role in the formation and early evolution of planetary systems (*1*, *2*). In protoplanetary disks (PPDs), large-scale magnetic fields threading the disk midplane are coupled to partially ionized gas and are thought to drive angular momentum transport and mass accretion onto the central star via magnetorotational instability and magnetized disk winds (*1*, *3*). Because the nebular magnetic field is sustained by the ionized gas in the disk, its presence or absence also serves as a proxy for the lifetime and distribution of the protoplanetary disk itself (*1*). While detection of magnetic fields in protoplanetary disks through observations provides information at scales of tens to hundreds of AU (*4*), magnetic fields recorded by meteorites can provide spatial and temporal constraints on the strength of the solar nebula field in the planet forming regions of PPDs (*1*, *5–10*). Paleomagnetic measurements of bulk meteorites and chondrules that acquired thermoremanent magnetizations in the early solar system nebula provide evidence for magnetic fields ranging from 54 to 100 μT within the first few million years (Myr) after the formation of calcium-aluminum-rich inclusions (CAIs) (*5*, *7*, *9*).

In addition to thermoremanent magnetization records, bulk paleomagnetic measurements of carbonaceous chondrites have explored the magnetic signatures of aqueously altered parent-body materials that carry chemical remanent magnetizations (CRMs) acquired when magnetic carriers, such as magnetite and pyrrhotite, precipitated from hydrothermal fluids in the presence of an ambient magnetic field (*6*, *10–13*). Because aqueous alteration in parent bodies of carbonaceous chondrites occurred ~3–5 Myr after CAI formation, straddling the expected epoch of nebula dissipation, CRM records from these materials may be used to determine if the nebular field was still present when aqueous alteration occurred (*14–16*).

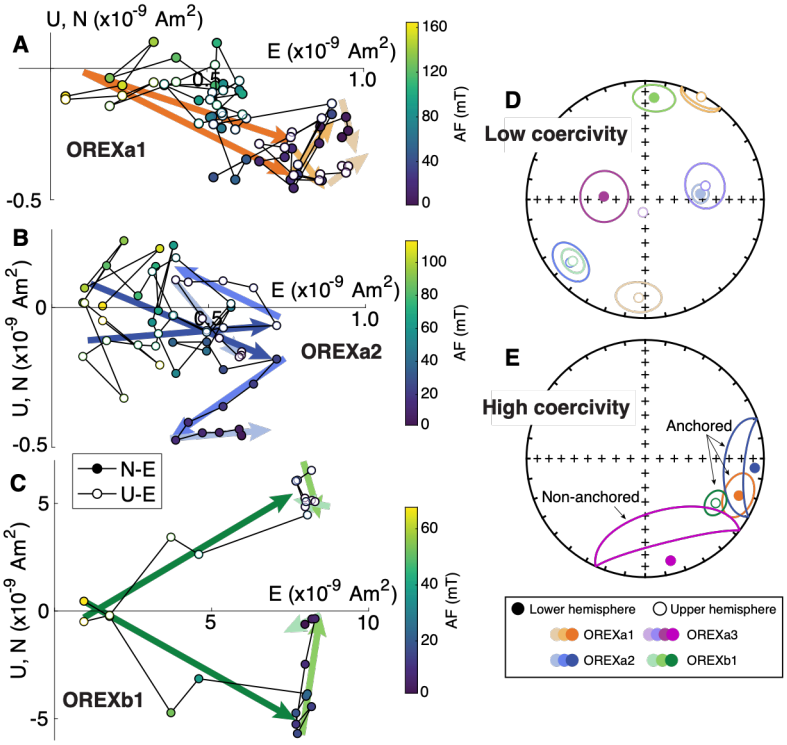

Fig. 1. AF demagnetization of NRM of selected Bennu subsamples. Orthographic projections of NRM vector endpoints during alternating field (AF) demagnetization for subsamples **(A)** OREXa1, **(B)** OREXa2 and **(C)** OREXb1. Closed symbols show the North (N)-East (E) projection of the magnetic moment and open symbols show Upper (U)-East projection of the magnetic moment. Color bars denote AF levels. Light colors arrow indicates the low coercivity components while the dark color arrow indicates the high coercivity component. Equal area stereographic projections of NRM component directions (circular markers) calculated from principal components analysis (PCA) and their corresponding maximum angular deviations (ellipses) are also shown for the **(D)** low and **(E)** high coercivity ranges of mutually oriented samples.

While CRMs recorded in meteorites have been used to reconstruct magnetic fields in the solar nebula, samples returned from asteroid 162173 Ryugu by JAXA's Hayabusa2 mission and asteroid 101955 Bennu by NASA's OSIRIS-REx mission have provided the first opportunity to evaluate this interpretation using returned asteroid material, enabling direct comparisons between the magnetic records preserved in meteorites and those retained by their parent bodies in the asteroid belt. Ryugu samples closely resemble CI chondrites in mineralogy, bulk chemistry, and isotopic composition, and contain framboidal magnetite and pyrrhotite formed during low-temperature aqueous alteration, making them potentially ideal CRM recorders (*12*, *13*, *17–22*). However, paleomagnetic analyses of Ryugu have yielded conflicting results with some samples reported to carry low coercivity stable natural remanent magnetizations (NRMs) interpreted as a solar nebular field, while others show no stable remanence along with rock magnetic evidence that Ryugu samples have poor magnetic recording properties, and suggest that prior work was potentially compromised by spacecraft magnetic contamination (*12*, *13*, *21*, *22*).

Here we report paleomagnetic and rock magnetic measurements of bulk samples returned from asteroid Bennu by NASA's OSIRIS-REx mission. Similar to Ryugu, Bennu's samples are dominated by hydrated phyllosilicates, magnetite, pyrrhotite, and carbonates, the mineralogical signature of aqueous alteration, and its parent body likely formed from the same heterogeneous reservoir in the outer protoplanetary disk to Ryugu and CI chondrites (*23–26*). The similarities between the returned materials allow us to probe the magnetic field present at the time and location when alteration occurred on the parent body in the outer solar system. Here, we report on the rock magnetic and paleomagnetic results of seven sub-samples from the parent sample OREX-800064-0.

Results

The samples' NRMs were measured using a 2G Enterprises superconducting rock magnetometer, and demagnetized using alternating field (AF) demagnetization with NRM component directions calculated using principal component analysis (*27*) (Fig. 1). Paleointensity estimates were acquired using the anhysteretic remanent magnetization (ARM) method (*6*, *7*, *28*) (see Methods). We also characterized the rock magnetic properties of two of the subsamples by obtaining hysteresis loops, First Order Reversal Curves (FORCs) and low temperature magnetic measurements (10–300 K magnetization and AC susceptibility), and by collecting Scanning Electron Microscopy (SEM) images (see Methods).

All samples showed the presence of low-coercivity components (up to ~15 mT) that are not clustered (Fig. 1D). We also observed the presence of multiple mid-coercivity components in all samples with coercivities ranging from 10 and 40 mT. All samples showed the presence of high-coercivity components that fully demagnetized between 70 and 160 mT (Fig. 1A-C; 1E). Among the seven targeted samples, four were mutually oriented, with a common high-coercivity magnetic direction between them (Fig. 1E).

We calculated a mean magnetic field intensity of 5.4 ± 1.7 μT (2σ uncertainty), assuming a correction factor for the ARM to the CRM (*f'*) of 3.33 to allow for a comparison with previously reported CRMs in the literature. Adopting a recently reported *f'* of 1.18, our measured paleointensity is of 15.3 ± 4.7 μT (2σ uncertainty) (*29*). Because the timescale for CRM acquisition in the parent body (10^5–10^6 years) exceeds the spin period of the parent body (several revolutions per day) (*30–35*), the components of the ambient magnetic field perpendicular to the spin axis time average to zero, and the CRM records only the field component parallel to the spin axis of the parent body (*7*). Given that the orientation of the spin axis relative to the solar nebula field is unconstrained, all spin axis orientations are assumed to be equally probable. A correction factor of 2 was included in the calculated paleointensity to take into account the full range of possible spin axis orientations (*7*), resulting in spin-corrected paleointensity of 10.8 ± 3.4 μT. Given the uncertainty associated with *f'* and the spin correction, we report a lower limit in the paleointensity acquired by Bennu's parent-body through a CRM of > 5.4 μT.

FORC diagrams from two subsamples show the presence of high coercivity, non-interacting magnetic recorders, indicated by a central ridge along the $B_u = 0$ axis that extends to microcoercivity (B_c) axis, suggesting the presence of values >100 mT consistent with robust single-domain magnetic carriers (Fig. 2A). In contrast, another subsample shows reduced B_c values and exhibits more vertical spread range of magnetic interactions along the B_u axis, typical of more multidomain particles (Fig. 2B). Low temperature measurements of these subsamples show the presence of the Verwey and Besnus transitions indicating the presence of magnetite and pyrrhotite in the subsamples (see SM). Overall, the two subsamples share the same magnetic mineralogy but differ in grain size, concentration, and degree of magnetic interactions. Backscattered electron (BSE) imaging and energy-dispersive X-ray spectroscopy (EDS) mapping (Fig. 2C and figs. S5-

S8) revealed the presence of fine-scale (nano- to micrometer) magnetite (Fig. 2D-F), mostly framboidal, and hexagonally shaped pyrrhotite grains (Fig. 2G-S). Generally, the magnetic mineral assemblage of Bennu is dominated by nm-µm magnetic carriers in the form of magnetite and pyrrhotite likely in the single-domain to single-vortex size range, that formed during parent body alteration and recorded a CRM in the Bennu parent body (*23–26*, *36–38*). No evidence of iron metal was observed in either magnetic or electron microscopy experiments.

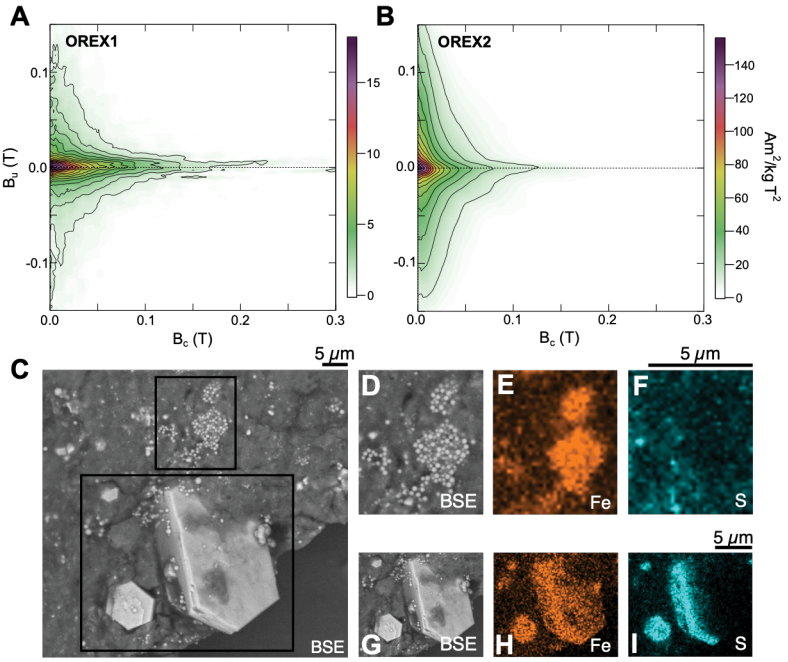

Fig. 2. Evidence of magnetic and pyrrhotite as magnetic carriers in the Bennu subsamples. First Order Reversal Curve (FORC) distributions are presented for subsamples **(A)** OREX1 and **(B)** OREX2, showing their magnetostatic interaction (B_u) and coercivity (B_c) fields. **(C)** Backscattered electron (BSE) image showing the presence of magnetite and pyrrhotite along with insets displaying elemental maps supporting the presence of **(D-F)** framboidal magnetite and **(G-I)** pyrrhotite.

Discussion

In the context of the meteorite dichotomy between non-carbonaceous (NC) and carbonaceous chondrites (CC) reservoirs (*39–42*), Bennu's parent body shares a common reservoir along with that of Ryugu and the CIs, implying a common location for these materials (*24*, *43*). Furthermore, Bennu likely formed in a region of the disk where ammonia ice was available (*25*), placing the formation region of Bennu beyond the water line, potentially >5 AU, assuming Sun-like parameters for modeling the location of ices in the early solar system (*44*, *45*). While ages for the aqueous alteration of Bennu are still needed, we adopt a recent estimate for the Ryugu CRM acquisition of ~3.5 Myr after the formation of the calcium-aluminum-rich-inclusions (CAI) as the age of the CRM of Bennu (*12*, *30*) due to the shared reservoir formation region. This age overlaps with those proposed for other carbonaceous chondrites that place an upper limit of the alteration of carbonaceous chondrites to ~5 Myr after CAI formation (*6*).

Our measurements also indicate that Bennu contains robust, high coercivity magnetic carriers. This is supported by the observation of framboidal magnetite and fine-grained (<2 µm) pyrrhotite among our subsamples (*38*, *46*, *47*). Framboidal magnetite is thought to form predominantly by single-phase precipitation from Fe-sulfide-bearing hydrothermal fluids (*11*, *37*, *48*, *49*). Critically, this single-phase precipitation mechanism produces magnetite with no inherited primordial Fe magnetization to bias the CRM acquisition, in contrast to isolated magnetite grains formed by pseudomorphic replacement of metallic Fe or sulfides that could be biased by the initial magnetization of the primordial iron observed in some carbonaceous chondrites (*6*, *10*, *11*). Pyrrhotite in CI chondrites are thought to form from a combination of the oxidation of troilite and precipitation during aqueous alteration (*50–53*). No indication of a pre-alteration magnetization

are observed in CI-like samples (e.g., associated with Fe-sulfides formed in the solar nebula), indicating that the majority of magnetic carriers in Bennu samples formed during alteration in the parent body (*52*).

If the magnetic record obtained from Bennu is placed beyond the CC reservoir (Fig. 3), the average magnetic field strength in each reservoir can be calculated and compared with predictions of magnetohydrodynamic (MHD) models that describe magnetically driven accretion in the early Solar System under magnetized disk winds (*1*, *5*, *7*, *9*, *10*, *12*, *13*, *22*, *54*). Previous studies identified evidence for magnetic heterogeneities in the early Solar System, particularly within the CC reservoir, relative to the predictions of magnetized disk-wind models (*5*, *54*). However, when considered alongside reported measurements, the reservoir-averaged field strengths exhibit an overall decrease with heliocentric distance, broadly consistent with magnetized disk wind model predictions (*1*, *5*, *12*). These results suggest that although local magnetic heterogeneities may have existed (*5*, *10*, *54*), the first-order trend was a progressive decline in magnetic field strength with increasing distance from the Sun.

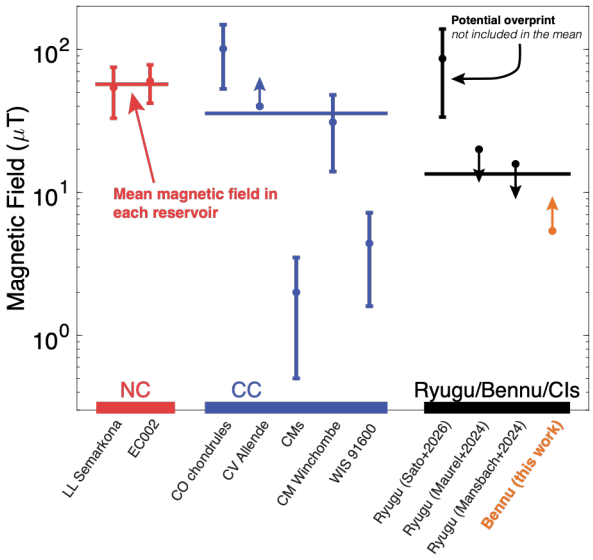

Fig. 3: Comparison Between Records of the Magnetic Field in the Early Solar System Across Different Reservoirs Show a Steady Decay of the Magnetic Field as a Function of Radial Distance. Data are from refs. [1, 5, 7, 9, 10, 12, 13, 22, 54].

When compared to the magnetic records reported for Ryugu samples (*12*, *13*, *21*, *22*), the paleomagnetic and rock magnetic results reported for Bennu here exhibits high-coercivity remanent magnetization that records ancient magnetic fields of a few microteslas, highlighting a striking contrast between the two asteroids and their parent bodies. Although the magnetite and pyrrhotite identified in samples from both bodies are morphologically similar, the fluids that mediated the formation and alteration of these magnetic carriers were likely compositionally distinct. In fact, the observation that Bennu samples contain a higher abundance of volatiles than Ryugu samples suggests that Bennu's magnetic mineral assemblage may have formed through a distinct aqueous alteration pathway influenced by volatile-rich fluids (*24*, *25*). Notably, the single-phase precipitation mechanism proposed to explain the morphology of magnetite in Bennu samples has been linked to the presence of volatiles such as ammonia, further supporting formation of the Bennu parent body beyond the nitrogen snowline in the early Solar System (*11*). While magnetite morphology is potentially important, our measurements suggest that not only the shape of the magnetic carriers but also the chemical composition of the alteration fluids may have influenced the acquisition of the CRM. That is because, although Ryugu samples also contain framboidal magnetite, these grains do not appear to be robust magnetic recorders, as indicated by the presence of low-coercivity magnetic components and by the upper limits derived for their solar nebula magnetic field records (*12*, *13*, *22*). This contrast may reflect differences in the aqueous alteration histories of the two parent bodies. For example, evidence suggests that the Bennu parent body experienced multiple episodes of aqueous alteration (*55*), which could have modified the concentration and composition of volatile species within alteration fluids. Variations in volatile abundance, driven by differences in local reservoir composition, parent-body formation location,

and/or the delivery of volatile-rich material, may therefore have played a key role in controlling the formation of magnetic carriers and ultimately the acquisition of magnetic records in the early Solar System. More broadly, our results emphasize that sample-return missions can provide novel results, revealing critical differences in alteration history, volatile evolution, and magnetic records, exposing a level of parent-body diversity that is not readily apparent from orbital measurements alone, even when targeting asteroids that appear compositionally similar.

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Competing interests: The authors declare that they have no competing interests.

Data, code, and materials availability: All data needed to evaluate the conclusions in the paper can be found here. Data will be available in the Magnetics Information Consortium (MagIC) Database upon publication.

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Supplementary Materials

Supplementary Text

Figs. S1 to S9

Tables S1 to S3

References *(1–43)*

1 Paleomagnetism

We obtained seven subsamples from the parent sample OREX-800064-0 using a diamond wire saw. Each subsample weighed ∼2-4 mg. Samples were mounted to non-magnetic quartz disks with cyanoacrylate glue for paleomagnetic measurements. Four samples were kept mutually oriented while the other three were not because they were too friable to handle. Paleomagnetic measurements were conducted at the Institute for Rock Magnetism (IRM) at the University of Minnesota and at the Massachusetts Institute of Technology (MIT). The samples' natural remanent magnetization (NRM) was measured using a 2G superconducting rock magnetometer, and demagnetized using alternating field (AF) demagnetization up to 169 mT with NRM component directions calculated using principal component analysis (*1*) (Table S1). If an NRM component not anchored to the origin had a deviation angle (DANG) less than the associated maximum angle deviation (MAD), then this component was inferred to be the characteristic component and a new PCA fit was computed anchored to the origin.

To obtain paleointensities, we used the anhysteretic remanent magnetization (ARM) method (*2, 3, 4*). For this type of experiment, we generally have:

$$B_{paleo} = \frac{B_{lab}}{f'} \frac{\Delta NRM}{\Delta ARM} \tag{1}$$

where B_{paleo} is the intensity of the ancient magnetic field recovered from the experiment, B_{lab} is the ARM bias magnetic field applied to the sample, ΔNRM and ΔARM are the respective changes in magnetic moment during the demagnetization of the NRM and the ARM, and in this case f' is the ratio of the efficiency of CRM to ARM (*5*). In our experiments, the ARM was applied with a peak AF of 169 mT and with a DC bias field of 50 µT. Paleointensity linear fits (reduced major axis or ordinary least squares) were selected following the procedures described in (*2*). Paleointensities were calculated for all identified components and are shown in Table S2 initially assuming an uncorrected f' (i.e. $f' = 1$). Fig. S1 shows the calculated paleointensities from the high coercivity ranges from all samples. We adopt two correction

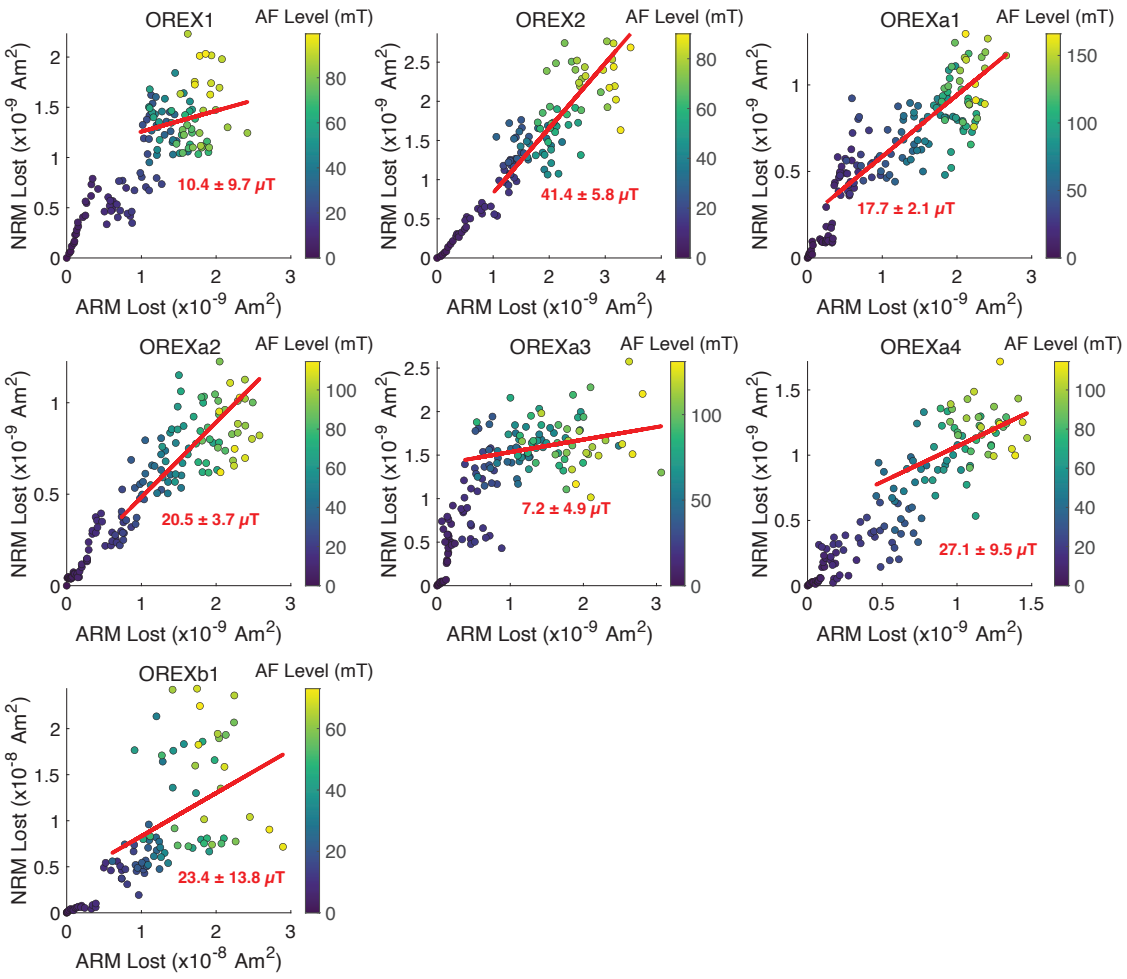

Figure S1: **Paleointensity experiments on Bennu samples showing the fitted HC components**. Shown is NRM lost versus ARM lost during AF demagnetization for each specimen. Red lines show the least squares fit over HC used to calculate the paleointensity. Paleointensities are listed in red with 95% confidence intervals. Color bars show the AF levels.

factors for our analysis, one is a f' from ARM to CRM of $f' = 3.33$ (*5*), used in previous publications of CRMs that assume magnetite as the main magnetic carrier, and the recent calculated $f' = 1.18$ based on magnetite growth experiments to reproduce CRM magnetic efficiency (*6*). We use only the anchored samples to calculate the main paleointensity reported in the main text (Table S3).

2 Rock Magnetic Measurements

We characterized the rock magnetic properties of two of the subsamples by obtaining hysteresis loops, backfield remanence data, first order reversal curves (FORCs), and low temperature magnetic measurements (10–300 K magnetization and AC susceptibility) at the IRM. We obtained Scanning Electron Microscopy (SEM) images using a Hitachi TM4000Plus at Purdue University. Together these analyses allow us to determine the magnetic properties of our samples, identify magnetic minerals present and assess the robustness of the observed magnetization.

2.1 Low-temperature magnetic properties

All low temperature magnetic measurements were conducted using a Quantum Design MPMS3 at the IRM. Cryogenic cooling experiments provide information about low-temperature magnetic transitions associated with specific magnetic minerals and can also be used to characterize magnetic domain state distributions. Room temperature saturation isothermal remanent magnetization (RTSIRM) experiments track the evolution of a 2.5 T IRM imparted at 300 K during thermal cycling from 300 K to 10 K and back to 300 K. Field-cooled (FC) experiments monitor how a sample's 2.5 T IRM (imparted at 10 K) evolves during warming from 10 to 300 K, after it has been cooled from 300 to 10 K in the presence of a 2.5 T field. Zero-field-cooled (ZFC) experiments monitor how a sample's 2.5 T IRM (imparted at 10 K) evolves during warming from 10 to 300 K, after it has been cooled from 300 to 10 K in a field free environment. The FC and ZFC experiments are particularly useful for characterizing samples containing magnetite, as the presence of a bias field during cooling across the Verwey transition ($\sim$120 K) can influence and limit the selection of crystallographic orientations of monoclinic magnetite below the transition. The ratio of FC to ZFC magnetization below the Verwey transition can therefore provide information about magnetic domain state.

In addition to these remanence-based experiments, we measured low-temperature ther-

momagnetic susceptibility at five different frequencies: 10.0, 26.6, 70.8, 188.2, and 499.5 Hz. During these measurements, the frequency of the applied field is varied, producing a delayed response in the magnetic system. As such, we measured both in-phase (κ') and quadrature (κ'') components of magnetic susceptibility, which provide additional information about the magnetic relaxation dynamics of constituent minerals.

The low-temperature remanence experiments reveal two characteristic magnetic transitions in both subsamples. A decrease in magnetization occurs in the ZFC and FC curves between 10 and ∼40 K (Fig. S2A, B), accompanied by an ∼2–10% loss in room-temperature saturation isothermal remanent magnetization (RTSIRM) (Fig. S2C, D). This RTSIRM behavior in particular is consistent with the Besnus transition of monoclinic pyrrhotite (*7*). Between 40 and 120 K, magnetization remains relatively stable, before undergoing a second pronounced decrease in both ZFC and FC curves, consistent with the Verwey transition of magnetite.

The FC/ZFC behavior below the Verwey transition provides further constraints on the magnetic domain state of the magnetite. In magnetite, the ratio of FC to ZFC magnetization at low temperatures (R_{LT}) depends on magnetic grain size, with single domain (SD) magnetite yielding $R_{LT} > 1$, multidomain (MD) magnetite yielding $R_{LT} < 1$, and pseudo single domain (PSD) magnetite falling near $R_{LT} \sim 1$ (*8, 9, 10*). In OREX1, $R_{LT} = 1.32$ at 20 K, suggesting predominantly SD behavior, whereas OREX2 has $R_{LT} = 0.97$, consistent with a greater contribution from larger grains. The morphology of the Verwey transition provides additional support for this interpretation. The sharp transition in OREX2 resembles that of synthetic MD magnetite with grain sizes > 1.5 μm (*8,10,11*), whereas the more gradual transition in OREX1 is characteristic of smaller SD-PSD grains (0.35–1.5 μm) (*12, 13*). These observations are consistent with the FORC measurements from these same samples (Fig. 2), which indicate higher coercivities in OREX1 and therefore a finer grain size distribution relative to OREX2.

The Verwey transition is also evident in the RTSIRM data (Fig. S2C, D). Samples de-

magnetize most rapidly during cooling between 150 and 120 K. During warming, both samples show a local magnetization minimum at ∼130 K, which is often observed in multidomain grains of magnetite. Following thermal cycling, the RTSIRM decreases by 22–42%. The greater loss observed in OREX2 is consistent with its greater contribution from MD magnetite, which is known to undergo substantial demagnetization across the Verwey transition (*14, 15*). Altogether, the low-temperature remanence experiments demonstrate the presence of both pyrrhotite and magnetite in the Bennu subsamples and reveal substantial differences in magnetic grain size between adjacent subsamples.

The low-temperature thermomagnetic susceptibility data provide independent support for the presence of both minerals while providing additional constraints on their magnetic domain states (Fig. S3). A local minimum at ∼30 K in the κ' data (Figs. S3A-D) corresponds to the Besnus transition of pyrrhotite (*7, 16*). However, the AC susceptibility measurements are particularly informative for magnetite as pyrrhotite does not exhibit significant frequency dependence in either κ' or κ'' (*17, 7*). The Verwey transition is visible at ∼120 K as an increase in κ' and a decrease in κ''.

The frequency dependence of κ' in subsample OREX1 (Fig. S3A) both above and below the Verwey transition, indicates a substantial fraction of magnetite near the superparamagnetic/SD threshold. As grains approach a stable SD phase, κ'' approaches zero (*18*), consistent with the order-of-magnitude weaker κ'' signal observed in OREX1. In contrast, the lack of a comparable frequency dependence in OREX2 (Fig. S3B) suggests a greater contribution from grains ranging from larger SD to MD configuration. This difference in magnetic grain size is further supported by the presence or absence of an inflection in the κ'' data at ∼40 K (MD). This low temperature anomaly in κ' is associated with electron hopping and domain wall relaxation in MD magnetites (*19, 20*) and is observed in OREX2 but not OREX1.

Overall, the low-temperature magnetic data indicate that (1) both subsamples are dominated by magnetite, with contributions from pyrrhotite; (2) OREX1 contains a smaller

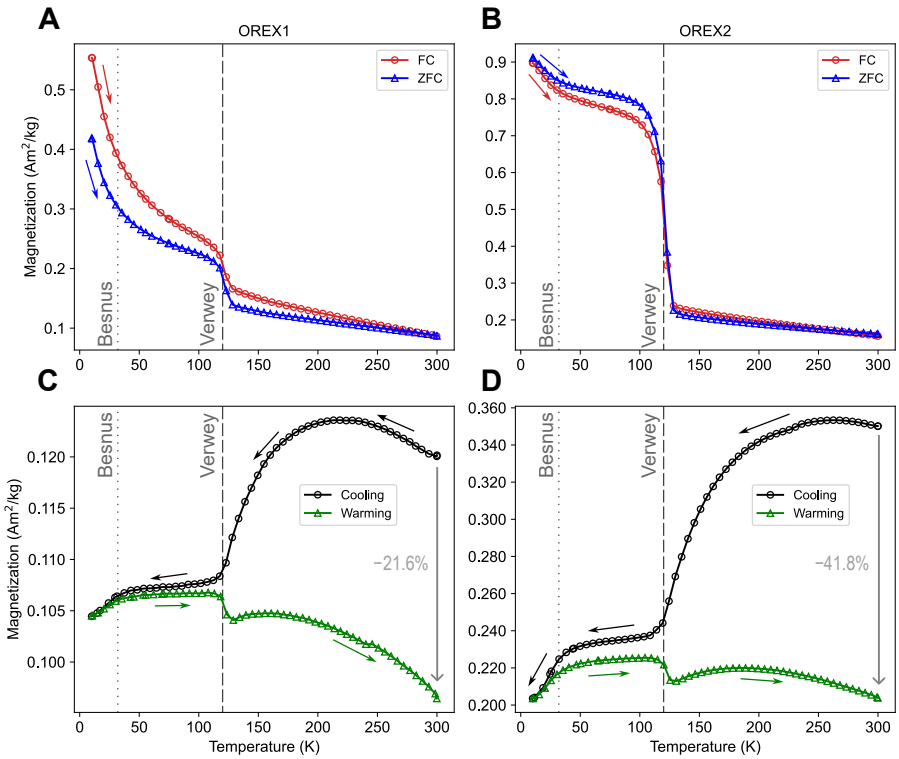

Figure S2: **Low-temperature remanence experiments.** Field-cooled (FC, red line with circles), zero-field-cooled (ZFC, blue line with triangles), and room-temperature saturation isothermal remanent magnetization (RTSIRM) curves for subsamples **(A, C)** OREX1 and **(B, D)** OREX2. Arrows next to each curve indicate the direction of measurement. The arrow between the cooling and warming RTSIRM curves indicates the magnetization lost during low-temperature demagnetization. Gray dotted and dashed vertical lines mark the diagnostic magnetic phase transition temperatures of pyrrhotite (Besnus transition) and magnetite (Verwey transition), respectively. Note: An instrumental artifact between 240–245 K in (C) was corrected for.

pyrrhotite contribution than OREX2; and (3) OREX1 contains finer-grained, more magnetically stable magnetite, whereas OREX2 contains greater proportion of larger MD magnetite. These results highlight the substantial magnetic heterogeneity that can occur between adjacent subsamples of Bennu material.

2.2 Bulk hysteresis analyses

Measuring bulk magnetic hysteresis loops and backfield magnetization curves and comparing hysteresis parameters allow us to assess differences in magnetic properties among samples

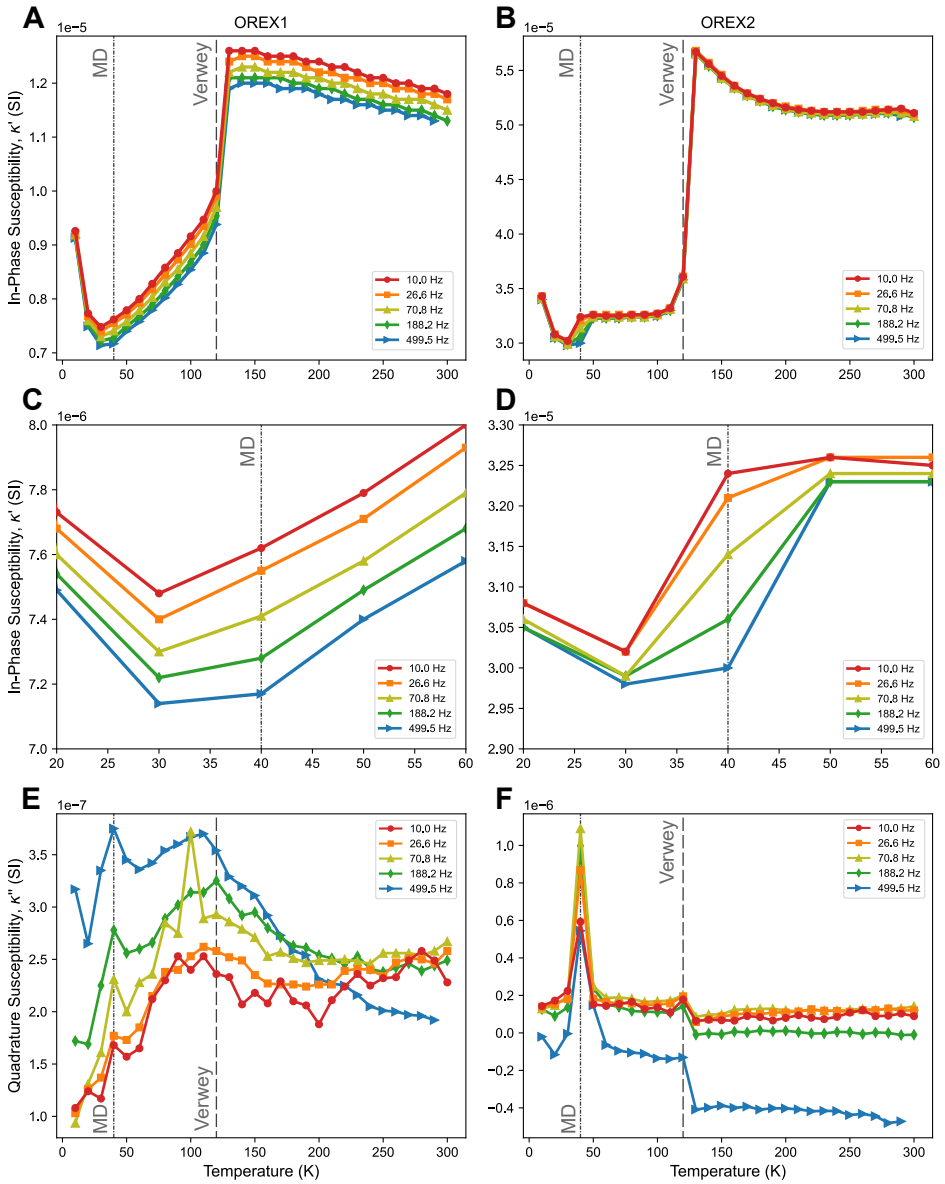

Figure S3: **Low-temperature complex AC susceptibility.** Temperature dependence of the in-phase (κ') and quadrature (κ'') components of AC susceptibility measured at frequencies ranging from 10.0 to 499.5 Hz for specimens **(A, E)** OREX1 and **(B, F)** OREX2. Panels **(C, D)** show enlarged views of panels **(A, B)**. Gray dash-dotted and dashed vertical lines indicate the diagnostic transition in multidomain magnetite (MD) and the Verwey transition, respectively.

to investigate potential heterogeneities between subsamples. All hysteresis measurements were collected using a Lake Shore 8600 vibrating sample magnetometer at the IRM. We observed that major hysteresis loops for subsamples OREX1 and OREX2 are narrow with relatively low coercivities (B_c) of 10.3 and 7.2 mT, respectively (Fig. S4A, B; red loops).

Mass-normalized saturation magnetization (M_s) and saturation remanence (M_r) in OREX2 are significantly higher than in OREX1, likely reflecting a higher concentration of magnetic minerals. Coercivity of remanence (B_{cr}) is higher in OREX1 (Fig. S4A, B; blue curves). The two subsamples have similar hysteresis ratios, with OREX1 having a slightly higher squareness ratio (M_r/M_s) and lower coercivity ratio (B_{cr}/B_c) than OREX2, suggesting more stable magnetic recording properties. We interpret the variation in M_s and M_r as representative of heterogeneity in magnetic carriers among subsamples, indicating that at the mm scale the composition and density of magnetic carriers varies, potentially reflecting how aqueous alteration influences the type and possibly shape of magnetic carriers in Bennu (*21, 22*). This heterogeneity is also observed in returned samples from Ryugu (*23*).

Coercivity unmixing was performed using MaxUnmix (*24*). The modeled coercivity spectra for both subsamples reveal two to three distinct magnetic components. The dominant component (orange curves, Fig. S4C, D) accounts for 72–73% of the total signal with coercivities of 58.7–66.7 mT, characteristic of fine-grained magnetite (*25*). In OREX2, this distribution is skewed right, likely representing a combination of fine-grained and coarser PSD-to-MD magnetite (*24*). A secondary component (green curves, Fig. S4C, D) contributes 18–23% of the signal with much lower coercivities (10.1–11.6 mT), consistent with large-grained pyrrhotite (*26*). A possible third component (red curves) shows anomalously high coercivities of 932.8 mT and 1012.1 mT for OREX1 and OREX2, respectively, and is most likely a high-field artifact caused by smoothing (*24*), inferred from: (1) negligible remaining magnetization at the end of demagnetization; (2) no corresponding high-coercivity signal in the FORC diagrams (Fig.2A, B); and (3) increased noise near magnetic saturation in the raw data. Taken together with the FORCs, low temperature magnetization data, and electron microscopy observations, the coercivity spectra indicate a magnetic mineral assemblage dominated by magnetite (∼76–80%) with minor contributions from large-grained pyrrhotite (∼20–24%).

We also summarize our data in a Day plot, which allows us to compare the bulk hysteresis

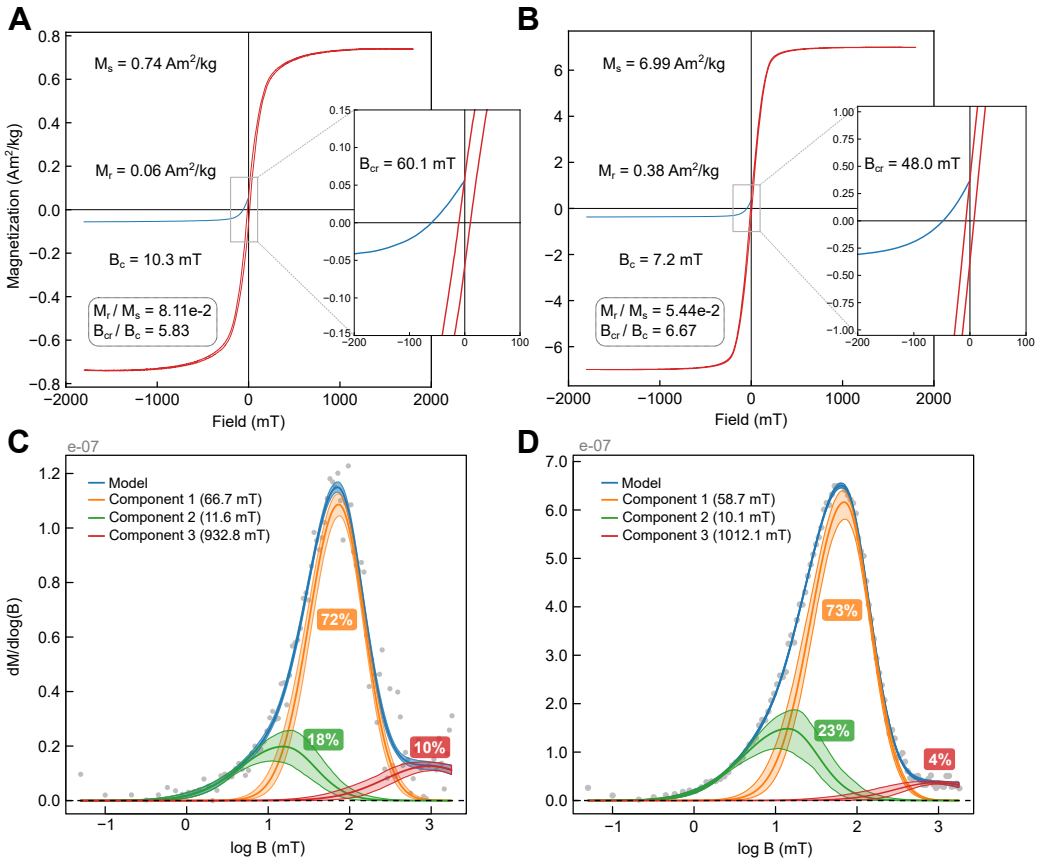

Figure S4: **Bulk hysteresis loops and modeled coercivity spectra for Bennu subsamples**. **(A, C)** OREX1 and **(B, D)** OREX2. The red loops and blue lines in (A, B) are processed hysteresis loops and backfield demagnetization curves, respectively. Hysteresis parameters saturation magnetization (M_s), saturation remanence (M_r), and coercivity (B_c) are denoted. The insets show scaled backfield curves from which the coercivity of remanence (B_{cr}) is obtained. Hysteresis ratios M_r/M_s and B_{cr}/B_c are noted in the lower left. In (C, D), coercivity data are shown by gray circles and low-, medium-, and high-coercivity components are color-coded green, orange, and red, respectively. The overall modeled coercivity spectra with 95% confidence band is shown in blue. Note: the high-coercivity component (red) is likely a high-field artifact.

parameters and the domain states of magnetic minerals of Bennu subsamples with other returned samples and chondrites (Fig. S5). In comparison to other chondrites, Bennu has bulk hysteresis ratios similar to CI chondrites and Ryugu samples, with coercivity ratios of 4-8 and squareness ratios 0.05-0.13 occupying a relatively small space in a Day Plot. This suggests relatively consistent magnetic mineralogies among CI-like bodies. Among CIs, Bennu is more similar to Ivuna and Tonk, whereas Ryugu is more similar to Orgueil and

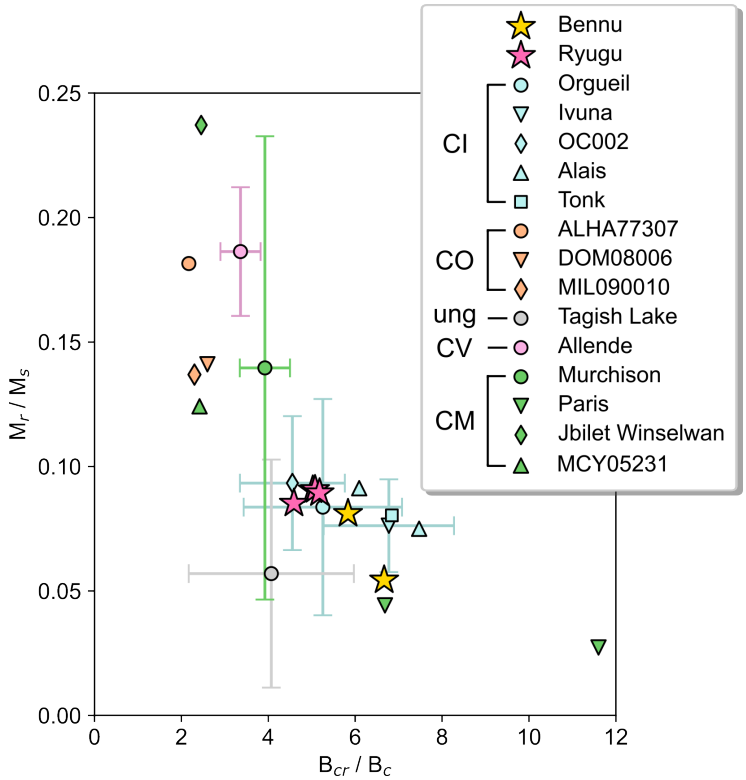

Figure S5: **Comparison to other chondrites.** Day Plot comparing bulk hysteresis parameters of Bennu (this study) with Ryugu and other chondrites. Averages with error bars (1σ) are given when three or more measurements are available. Other data are from: Ryugu (*30,31*), Orgueil (*31,32,33,34*), Ivuna (*33,34*), Oued Chebeika 002 ("OC002") (*34*), Alais (*34*), Tonk (*34*), DOM08006 (*33*), MIL090010 (*33*), Tagish Lake (*31, 32*), Allende (*32, 35, 36*), Murchison (*32, 33*), Paris (*33, 37*), Jbilet Winselwan (*33*), and MCY05231 (*33*). ALHA77307 data measured in this study.

Oued Chebeika 002. CM chondrites exhibit much more variability, likely due to their more diverse magnetic mineralogies with varying proportions of magnetite and sulfides as well as higher concentrations of metal and schreibersite (*27, 28, 29*).

2.3 First Order Reversal Curve (FORC) distributions

FORC diagrams can be used to identify the domain state of the magnetic minerals in our subsamples. FORCs (N = 220) were obtained with a maximum B_c field of 300 mT and B_u fields ranging from -300-200mT. FORC diagrams were processed using the FORCinel software following the VARIFORC protocol (*38, 39*). Due to the long duration (∼5 hours) measurements, datasets were drift corrected. First and last points were replaced and lower

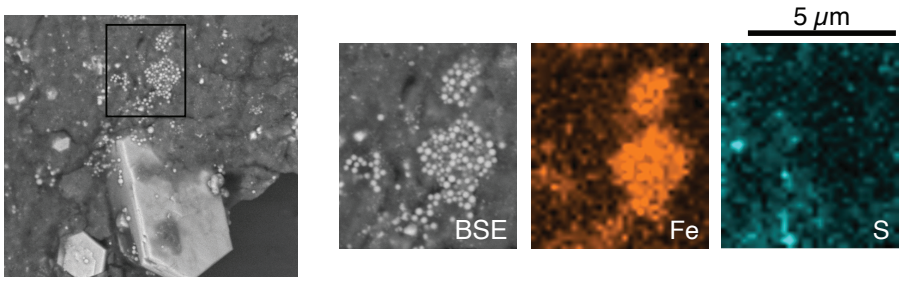

Figure S6: **BSE and EDS images of framboidal magnetite in our subsamples.** We observe a range of sizes, with most of the framboids being <0.5 µm.

branches were subtracted. VARIFORC smoothing factors for the vertical ridge ($S_{c,0}$), horizontal ($S_{c,1}$), central ridge ($S_{b,0}$), and vertical ($S_{b,1}$) were assigned as: $S_{c,0} = 7$, $S_{c,1} = 7$, $S_{b,0} = 5$, and $S_{b,1} = 7$ (OREX1) and $S_{c,0} = 5$, $S_{c,1} = 7$, $S_{b,0} = 5$, and $S_{b,1} = 7$ (OREX2). A horizontal and vertical lambda value of 0.1 was used for both subsamples.

Figs. 2A-B in the main manuscript show FORC distributions obtained for OREX1 and OREX2. While the FORC diagram for OREX1 shows the presence of a central elongated ridge associated with non-interacting, single-domain/single-vortex particles, the diagram for OREX2 shows a widened interactions along the B_u axis, supporting the presence of larger, multidomain particles (*40, 41*). Overall, OREX1 is likely to contain more stable magnetic particles than OREX2, further supporting the presence of magnetic heterogeneity among our subsamples.

2.4 Imaging subsamples

We obtained backscattered electron (BSE) images along with energy dispersive X-ray spectroscopy (EDS) maps to visualize the size, shape and composition of the magnetic carriers present in Bennu subsamples. These images were obtained from unpolished fragments remaining after cutting the subsamples used for our paleomagnetic and rock magnetic measurements described above. Fig. 2C-I from the main text shows a typical observation of magnetic carriers in Bennu. There are iron oxides, likely magnetite supported by the observed Verwey transition in our subsamples, and iron sulfides, likely pyrrhotite supported

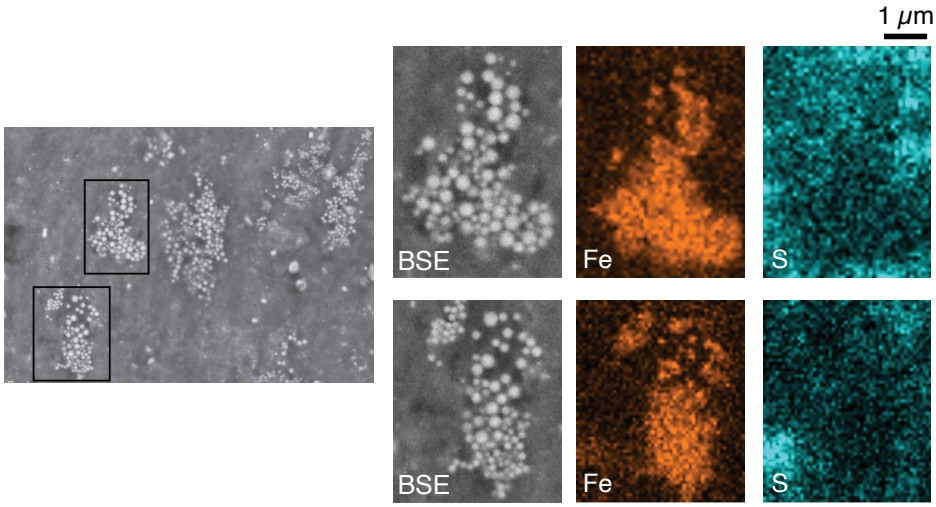

Figure S7: **Additional BSE and EDS images of framboidal magnetite in our subsamples.** We observe a range of sizes, with most of the framboids being <0.5 µm.

by the observed Besnus transition. Magnetite usually occurs in the form of framboids, but also as plaquette and radial shapes (*42*). We observe magnetite with a range of sizes, mostly under 0.5 µm (Figs. S6 and S7). Magnetite framboids occur both as dense clusters, but also as more finely disseminated grains in the groundmass of the material. The observed pyrrhotite is hexagonal in shape, with sizes ranging from µm to several tens of µm (Fig. S8). Magnetite and pyrrhotite appear isolated, but in some cases framboids are observed on the surface of pyrrhotite grains (Fig. S9). We observe a higher density of magnetite grains that are small (<0.5 µm) and more likely to explain the central ridge observed in the FORC (Fig. 2) associated with single-domain carriers (*40,41*). Nonetheless, the observed range of sizes of pyrrhotite grains allow for some of them (<1 µm) to be in single-domain state (*43*). Overall, magnetite is likely to be the main magnetic carrier in the form of single-domain grains, but we cannot rule out pyrrhotite as a potential magnetic carrier in the same stability range.

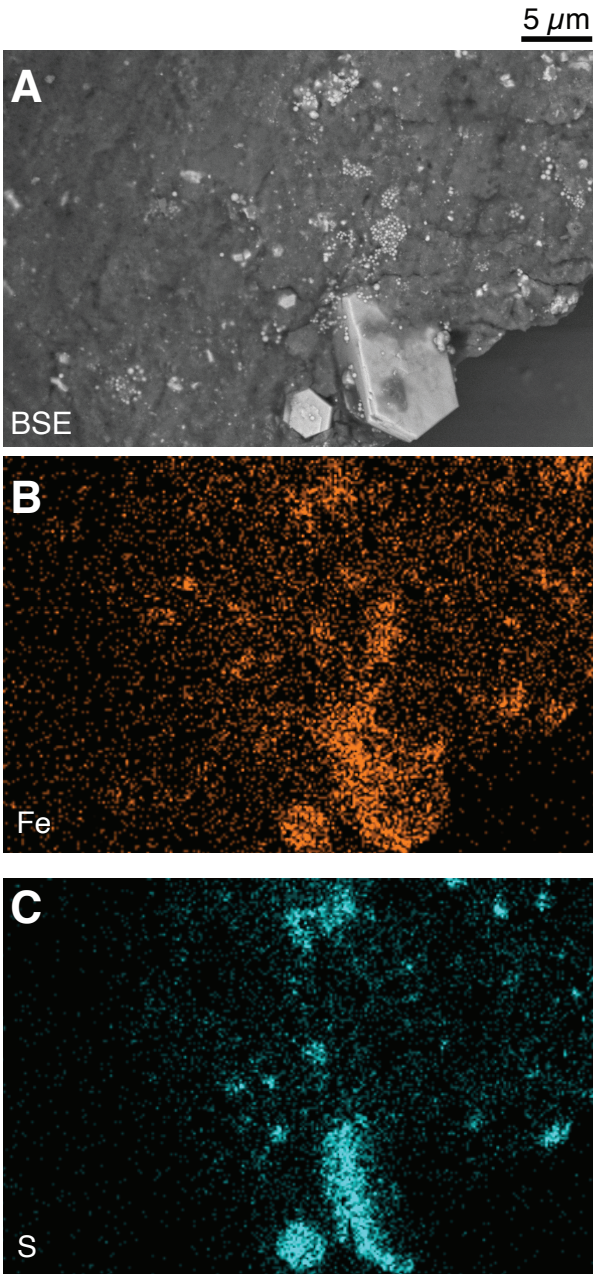

Figure S8: **BSE and EDS overview of magnetite and pyrrhotite grains in subsamples.** Zoom out image of Fig. 2 showing the distribution of sizes and shapes of magnetite and pyrrhotite grains found in Bennu. Not all magnetite grains are framboids, although they appear to be the most common shape of magnetite in Bennu samples.

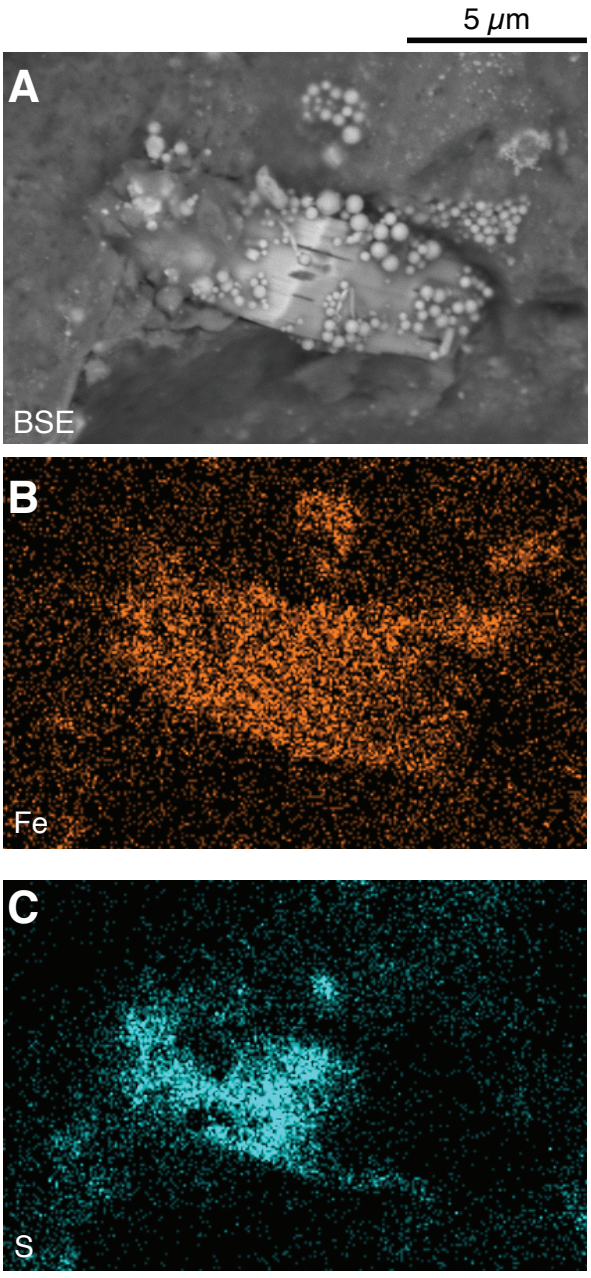

Figure S9: **BSE and EDS maps showing a association of magnetite and pyrrhotite grains.** Not all magnetite grains are associated with these iron sulfides. The BSE image of the pyrrhotite grain shows evidence of fine scale layering within the sulfide's basal plane that is consistent with intergrowths of hexagonal and monoclinic pyrrhotite.

Table S1: **Summary of components from AF demagnetization identified using principal component analysis (PCA).** The first column lists the sample name, the second column lists the component (LC for low coercivity and HC for high coercivity), the third column lists the number of AF steps used to calculate the mean, the fourth column lists the AF range used to calculate the PCA fit in mT, the fifth, sixth, and seventh columns show the results of the PCA [declination, inclination, and maximum angle deviation (MAD)], the eighth column lists the deviation angle (DANG) for the component, the ninth column denotes whether the fit is anchored (Y for yes and N for no), and the tenth column lists the MAD when the fit is not anchored.

Sample	**Comp.**	**Steps**	**Range (mT)**	**Dec. (°)**	**Inc. (°)**	**MAD (°)**	**DANG (°)**	**Anc.?**	**MAD (°) not anch.**
OREX1	LC1	5	1.4–9	177.5	+7.2	20.1	–	N	–
	LC2	5	11.5–21.5	96.9	+71.4	35.3	–	N	–
	HC	5	21.5–99.5	194.4	+8.8	20.5	22.5	Y	36.7
OREX2	LC1	10	2.7–12.75	188.2	−37.4	6.5	–	N	–
	HC	10	12.75–118.25	23.5	−44.7	23.6	58.2	N	–
OREXa1	LC1	5	1.4–6.5	183.8	−18.1	12.7	–	N	–
	LC2	5	6.5–14	28.6	−1.6	10.7	–	N	–
	HC	5	14–166	111.6	+16.2	12.6	17.4	Y	28.4
OREXa2	LC1	5	1.4–14	84.1	+51.4	6.0	–	N	–
	LC2	5	14–24	229.9	−18.6	12.4	–	N	–
	HC	5	24–114.5	94.9	+8.1	25.7	38.1	Y	38.7
OREXa3	LC1	5	1.4–9	190.0	−81.1	–	–	N	–
	LC2	5	9–19	77.2	−46.8	14.5	–	N	–
	LC3	5	19–24	274.2	+61.4	17.0	–	N	–
	HC	5	24–131	166.0	+12.3	40.3	66.7	N	–
OREXa4	LC1	5	1.4–9	15.1	+30.3	9.6	–	N	–
	LC2	5	9–21.5	225.5	−36.6	8.5	–	N	–
	LC3	5	21.5–28	60.1	+34.8	3.9	–	N	–
	LC4	5	28–38	311.2	+4.5	10.8	–	N	–
	HC	5	38–114.5	188.8	−15.5	24.1	45.8	N	–
OREXb1	LC1	5	1.4–6.5	229.6	−20.9	7.7	–	N	–
	LC2	5	6.5–16.5	5.0	+14.9	11.5	–	N	–
	HC	5	16.5–73	122.3	−30.4	7.9	7.8	Y	14.2

Table S2: **Summary of paleointensities from AF demagnetization of all samples (All).** The first column lists the sample name, the second column lists the component (LC for low coercivity and HC for high coercivity), the third column lists the number of AF steps used to calculate the mean, the fourth column lists the AF range used to calculate the fit in mT, the fifth and sixth columns show the calculated paleointensity and the associated error. The bottom rows report the mean paleointensity and standard deviation (1σ and 2σ) of the sample mean across all samples.

Sample	**Component**	**Steps**	**Range (mT)**	**Paleointensity (μT)**	**Error (μT)**
OREX1	LC1	5	1.4–9	109.1	10.4
	LC2	5	11.5–21.5	49.8	13.6
	HC	5	21.5–99.5	10.4	9.7
OREX2	LC1	10	2.7–12.75	41.4	4.0
	HC	10	12.75–90	41.4	5.8
OREXa1	LC1	5	1.4–6.5	29.0	11.2
	LC2	5	6.5–14	45.3	48.9
	HC	5	14–166	17.7	2.1
OREXa2	LC1	5	1.4–14	40.1	7.3
	LC2	5	14–24	54.1	13.9
	HC	5	24–114.5	20.5	3.7
OREXa3	LC1	5	1.4–9	139.2	31.4
	LC2	5	9–19	49.3	34.4
	LC3	5	19–24	16.8	61.8
	HC	5	24–131	7.2	4.9
OREXa4	LC1	5	1.4–9	71.2	24.2
	LC2	5	9–21.5	52.3	37.0
	LC3	5	21.5–28	104.3	180.2
	LC4	5	28–38	5.4	67.2
	HC	5	38–114.5	27.1	9.5
OREXb1	LC1	5	1.4–6.5	16.8	3.7
	LC2	5	6.5–16.5	49.2	17.3
	HC	5	16.5–73	23.4	13.8

200

Table S3: **Summary of high-coercivity paleointensities from AF demagnetization.** The first column lists the sample name, the second column lists the AF range used to calculate the fit in mT, the third and fourth columns show the calculated paleointensity and its error, and the fifth column denotes whether the fit is anchored (Y for yes and N for no). In bold are highlighted the paleointensities used to calculate the mean field reported in the main text. The corrected paleointensity accounts for a factor of 2 correction applied to the mean.

Sample	**Range (mT)**	**Paleointensity (μT)**	**Error (μT)**
OREX1*	**21.5–99.5**	**10.4**	**9.7**
OREX2	12.75–90	41.4	5.8
OREXa1*	**14–166**	**17.7**	**2.1**
OREXa2*	**24–114.5**	**20.5**	**3.7**
OREXa3	24–131	7.2	4.9
OREXa4	38–114.5	27.1	9.5
OREXb1*	**16.5–73**	**23.4**	**13.8**
		Paleointensity (μT)	**Error 2σ(μT)**
Mean		**21.1**	**8.6**
Mean (*anchored only)		**18.0**	**5.6**
Mean ($f' = 3.33$, *anchored only)		**5.4**	**1.7**
Mean (rotation, $f' = 3.33$, *anchored only)		**10.8**	**3.4**
Mean ($f' = 1.18$, *anchored only)		**15.3**	**4.7**
Mean (rotation, $f' = 1.18$, *anchored only)		**30.6**	**9.4**

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